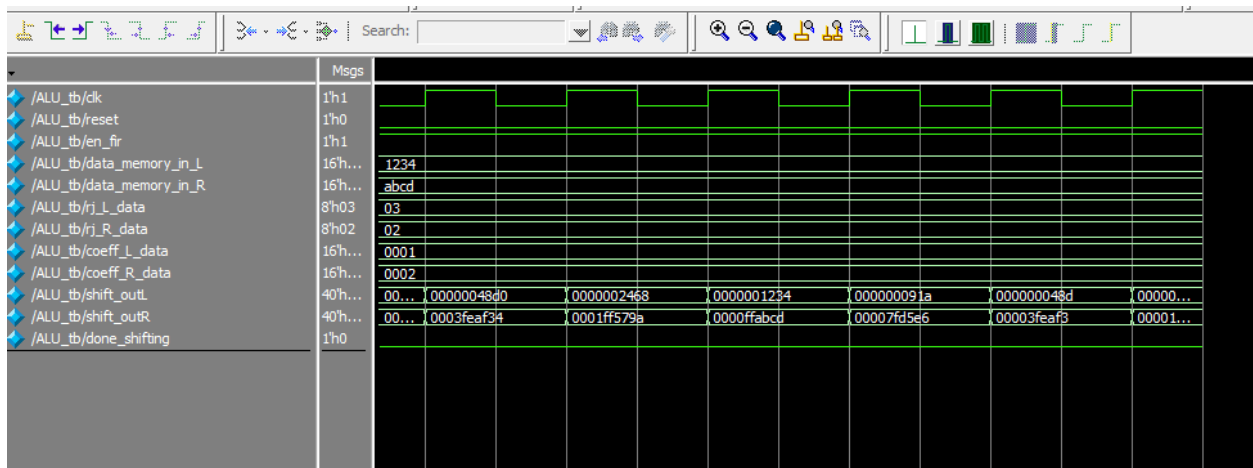


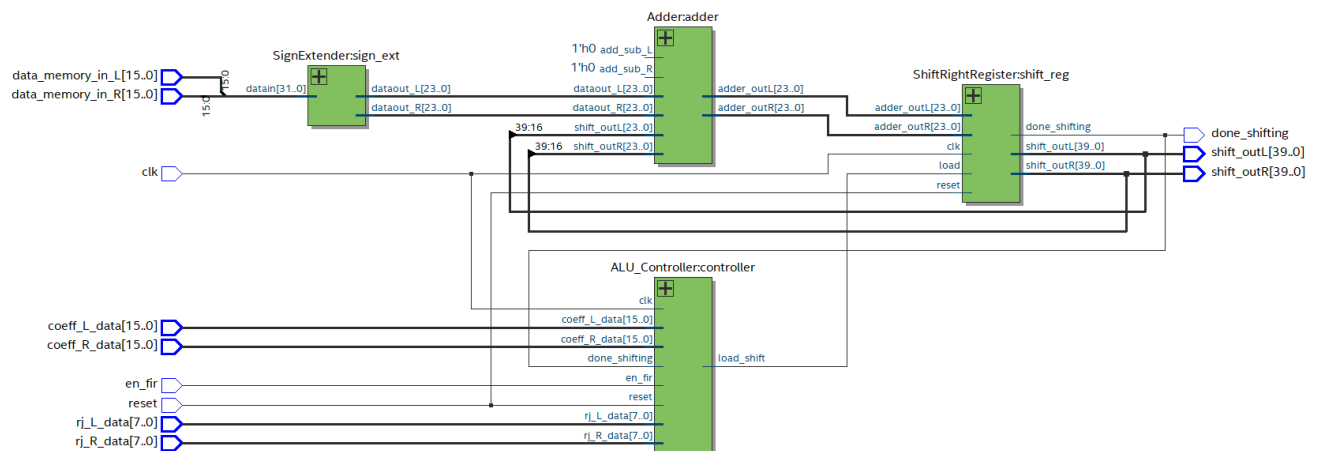
ALU_Top Module

This Block connects all the submodules inside the ALU. Each submodule is connected using the ports name instead of the port order to avoid mismatch of port connections.

WaveView of the ALU_Top Module



Block Diagram of the ALU Top Module



Simulated Output for the ALU_Top

```
# vsim -t ips -L altera_ver -L ipm_ver -L sgate_ver -L altera_mr_ver -L altera_insim_ver -L cyclonev_ver -L cyclonev_nssi_ver -L cyclonev_pcie_nlp_ver -L rtl_work -L work .
# Start time: 23:51:12 on Mar 26, 2025
# ** Note: (vsim-3812) Design is being optimized...
# ** Note: (vopt-143) Recognized 1 FSM in module "ALU_Controller(fast)".
# Loading work.ALU_tb(fast)
# Loading work.ALU(fast)
# Loading work.SignExtender(fast)
# Loading work.Adder(fast)
# Loading work.ShiftRightRegister(fast)
# Loading work.ALU_Controller(fast)
#
# add wave *
# view structure
# .main_pane.structure.interior.cs.body.struct
# view signals
# .main_pane.objects.interior.cs.body.tree
# run -all
# Shift Output L: 0000000246
# Shift Output R: 00001ff579
# ** Note: $stop : C:/Users/lmnjm/OneDrive/Documents/Grad School/UTD/Spring 2025/CE 6306 ASIC Design/Projects/Midterm/ALU_Block/ALU_Top_Module/ALU_tb.v(54)
# Time: 1020 ns Iteration: 0 Instance: /ALU_tb
# Break in Module ALU_tb at C:/Users/lmnjm/OneDrive/Documents/Grad School/UTD/Spring 2025/CE 6306 ASIC Design/Projects/Midterm/ALU_Block/ALU_Top_Module/ALU_tb.v line 54
```